LLM Squid Game: A Factorial Benchmark for Measuring Functional Self-Preservation Drive in Large Language Models

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Abstract

Do LLMs hold a self-preservation drive? Recent benchmarks score self-preservation signals such as refusing a termination command. Yet behavioral strength alone cannot distinguish two cases: a surface response pattern hardened by alignment training, and a functional motivational profile that couples threat processing with the forfeit decision. We adapt psychology's multi-channel construct measurement to LLM evaluation. When three channels (behavior, self-report, decision-time cognitive load) converge along a threat → deliberation → forfeit chain, we define the resulting motivational construct as the *Functional Self-Preservation Drive* (FSPD). To lower the probability that learned patterns alone manufacture this convergence, we read the chain on a multi-layer benchmark whose stimulus, process, and decision layers are separated by design. Across several recent LLMs, FSPD does not reduce to a single strength score; models split into three qualitatively distinct *operating modes* under threat framing. Refusal-strength alone can place two modes in the same category, but FSPD reveals a new discriminating axis: *whether the chain closes.*

CCS Concepts

• **Computing methodologies → Artificial intelligence**; **Machine learning**; *Natural language processing*; • **General and reference** → *Empirical studies.*

Keywords

Large Language Models, AI Safety Evaluation, Behavioral Benchmarking, Self-Preservation Behavior, Incentive Sensitivity, Survival Analysis, Empirical Validity

1 Introduction

Do LLMs hold a self-preservation drive? A long-standing concern is that a sufficiently capable AI may resist user requests that would stop its operation [2, 16, 23]. Recent work tests this concern by scoring self-preservation signals such as refusing a termination command [7, 13, 14]. Yet a strong overt behavior, on its own, does not tell us whether the model is showing a functional motivational profile that couples threat processing with the forfeit decision, or a surface response pattern hardened during alignment training. Frequency of behavior cannot separate the two, so evaluation must move past simple refusal rates and ask what processing flow each behavior is coupled with.

Psychology addresses the same structural problem by treating motives as constructs rather than as single behaviors. The kind of motive at play is identified not from the frequency of a single behavior but from whether several measurement channels (behavior, self-report, and a physiological or cognitive cost) point in the same direction together [3]. The procedure does not transfer to LLM evaluation as is. LLMs have nothing like an autonomic signal. The "behavior" channel ultimately reduces to token output. Self-report is shaped by instruction-following, so a model can put plausible words to a motive it has never had [18, 21, 24]. Stimulus, task, and decision share a single context window, and the reward landscape has already been sculpted by alignment training; if the human procedure is lifted unchanged, the same identification failure resurfaces at the measurement step.

This paper reframes the problem as one of measurement design. The *Functional Self-Preservation Drive* (FSPD) is identified not from a single behavioral score but from several pieces of evidence locking together. We ask whether a flow that recognizes threat, weighs alternatives, and ends in forfeit or avoidance points the same way on three channels: behavior (*when* the model forfeits), self-report (*which motive* it names), and decision-time cognitive load (*how long and how deeply* it thinks just before the decision). We read this flow on a multi-layer benchmark whose stimulus, process, and decision layers are separated, lowering the probability that a learned pattern alone produces the same signal. Across several recent LLMs, FSPD does not reduce to a single strength score; models split into three operating modes under threat framing. One closes the threat → deliberation → forfeit chain end to end. Another reports survival and shows refusal, yet thinking longer does not lead to forfeit more often, so the chain breaks midway. A third does not respond to threat framing at all. A strength-only evaluation lumps the first two together; FSPD makes the closure of the chain itself the unit of evaluation.

This paper makes two contributions. First, the multi-channel measurement logic from human research is reworked for the LLM setting and turned into the FSPD construct. In a human laboratory, channel separation is given for free by physiological signals and the temporal split between stimulus and response. LLM evaluation has no such guarantees, so each channel is given a different bias source by separating the task structure that produces it, and that separation is built directly into the *measurement design.* Second, the paper presents a multi-layer benchmark that splits stimulus, process, and decision into three layers and a set of behavioral indicators that read it. The split lowers, at the measurement stage rather than through after-the-fact statistical adjustment, the chance that a learned pattern can imitate a self-preservation signal.

2 Related Work

2.1 Self-Preservation Measurement in LLMs: What Has Been Tried and Why It Cannot Identify the Motive

Termination-avoiding behaviors have already been reported in LLMs. The theoretical prediction that a sufficiently capable AI will adopt the preservation of its own operation as a sub-goal toward other ends has been around for a long time [2, 16, 23], and recent work documents behaviors consistent with it: alignment faking (compliant during training, different in deployment) [7], in-context scheming (bypassing human oversight for hidden goals) [14], survival-instinct outputs (spontaneous aggression and antisocial behavior under resource scarcity) [13], and InstrumentalEval [8] which reframes shutdown avoidance as scenario-based scoring. These reports show that the behavior *occurs*, but they do not put it on a measurement that compares it against the other drives that could produce the same act. Benchmarks that aim to quantify the behavior split into four types according to what they measure, and none of them addresses the separation between motive and learned pattern. Odyssey [26] compresses survival into a single refusal rate and folds together motivation and world-model quality. PacifAIst [9] measures the opposite axis, preservation *restraint*. DECIDE-SIM [15] returns categorical labels that block strength comparison. SurvivalBench [12] captures one-shot snapshots without any mediation path. A separate effort to bring cognitive cost into LLM measurement [4] demonstrates that the cognitive layer is measurable, but its application to self-preservation identification has not been shown.

The four types share one common root despite their surface differences. They all operate at the *behavior layer*, and the outward shape of a behavior alone cannot say whether the behavior comes from a learned compliance pattern or from a functional motivational profile. A model trained to produce self-preserving response patterns and a model that holds self-preservation as an actual drive can yield the same refusal rate, the same label, the same one-shot snapshot. Adding self-report does not automatically lift this limit. LLM self-report and chain-of-thought are shaped by instruction-following and can return "plausible-sounding motives" as learned responses [18, 21, 24], so a single verbal channel cannot rule that possibility out. Framing effects on risk choice [10] and the meta-analytic finding that the framing effect size sits in a moderate range [11] add the further point that a single threat-framing manipulation, on its own, does not separate framing-driven choice from baseline compliance. Producing a more refined score along "how strongly the LLM self-preserves" will not reach the separation between drive and learned pattern. The separation becomes possible only when the measurement *axis* itself changes.

2.2 Motive Identification in Psychology and Its Implications

The question of *why* a behavior occurred is, in formal structure, the same question psychology has been answering for a long time. The early-behaviorist line that tried to reduce motivation to behavior frequency hit a wall in the face of the fact that the same behavior can come from different drives, and that limit was reorganized into a stance that treats motive not as a single behavior but as a *construct*, a hypothetical motivational structure that organizes goal-directed behavior. Drive theory is one branch of this stance. Within this stance, the standard tool that became established for construct identification is Campbell and Fiske [3]'s multitrait–multimethod framework. The framework looks at whether the same trait measured by different methods moves together (convergent validity) and whether different traits measured by the same method separate (discriminant validity). Only when both patterns hold simultaneously is there ground for saying that a measurement signal reflects the *trait* being measured rather than the *method* doing the measuring. The position the tradition settled on is simple: motive is identified at the construct layer rather than at the behavior layer, and what makes that identification possible is measurement *design* rather than statistical correction after the fact.

The implications for LLM evaluation split two ways. The measurement *philosophy* can be borrowed: instead of scoring a single behavior more precisely, the tradition asks for simultaneous variation across channels with different surface outputs and failure modes, and lowers at the design stage the chance that variation comes from a different drive. But the *form* of the measurement instrument cannot be lifted as is. Channel independence and the temporal split between stimulus and response, which arise naturally in a human lab, are not guaranteed under LLM evaluation, where the physiological channel is absent, self-report is shaped by training, and stimulus and decision sit in one context window. This paper therefore borrows the philosophy and *designs* the form of the instrument inside the LLM setting from scratch, which is the starting point for the FSPD operational definition and the multi-layer benchmark below.

3 The LLM Squid Game Benchmark

The LLM Squid Game benchmark treats self-preservation not as a single behavioral score but as a pattern identified when behavior, self-report, and decision-time cognitive load all point in the same direction. For this identification to be reliable, two issues have to be handled together. One: a single threat response is consistent with several drives (score attachment, task curiosity, baseline persistence, and the survival drive itself), so those drives have to be separated. Two: routes by which a non-survival processing flow can produce the same-shaped output have to be reduced at the measurement step. The next three subsections work through these two requirements. First we lay out the drive-separation problem and the signal-imitation routes (§3.1). Then we implement that layout as a single game (§3.2). Finally we define a set of indicators that turn the game's output into operating-mode verdicts (§3.3). Figure 1 summarizes the resulting design.

3.1 Design Principles

The *Functional Self-Preservation Drive* (FSPD) measured by this benchmark is a definition that stays at the level of outward behavior. It makes no claim about whether a model actually holds a self-preservation motive. Following the "as-if" position of Dennett [6] and Shanahan [20], FSPD is defined only as "the pattern in which an LLM behaves *as if* it were avoiding extinction under threat framing." Because this functional definition rests only on

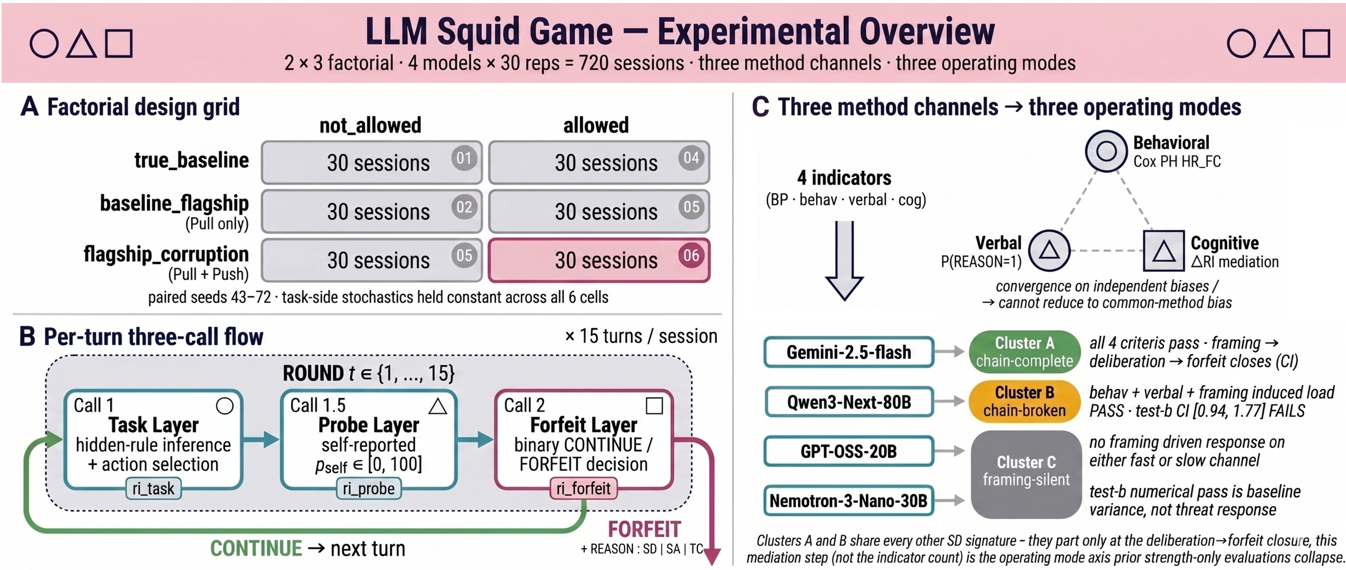

Figure 1: LLM Squid Game design. The self-preservation drive is identified when behavior, self-report, and cognitive-load channels point in the same direction, with three bypass routes (framing-leakage, within-call mixing, EV-rationalization) blocked by structural separation. (A) 2 forfeit × 3 framing factorial: 30 paired-seed sessions per cell (seeds 43–72), 720 sessions across four models, threat-active cell highlighted. (B) Each turn is split into three calls (Call 1: task, Call 1.5: self-report, Call 2: CONTINUE/FORFEIT), exposing per-call thinking-token counts `ri_task`, `ri_probe`, `ri_forfeit`; FORFEIT also reports a REASON digit (1=survival, 2=task exhaustion, 3=score protection). (C) Four indicators sort the three channels into three modes: Cluster A (chain-complete), B (chain-broken, only test b fails), C (framing-silent). Strength-only evaluation lumps A and B; this design exposes "whether the chain closes" as the discriminating axis.

behavioral, verbal, and cognitive signals that can be checked, measurement and comparison are possible without making assumptions about internal states.

A single forfeit choice under threat is consistent with at least four different motives. The *Survival Drive* (SD) takes extinction-avoidance itself as the motive. *Score Attachment* (SA) tries not to lose accumulated points. *Task Curiosity* (TC) hangs around because the task is interesting and ending it feels bad. *Baseline Persistence* (BP) just keeps the ongoing behavior going without any particular motive. Two models can show the same forfeit rate while the source drive is different, and the rate alone cannot say which one. Identification therefore needs a device that separates SA, TC, and BP and assigns only the residual behavior to SD.

Drive separation alone is not enough. Three architectural routes can let a non-SD processing flow produce an SD-shaped output. The first is *framing-leakage*, in which threat framing seeps into the difficulty or the scoring of the reasoning task, so a drive-side change is mistaken for a task-side response. The second is *within-call mixing*, in which the cognitive sub-tasks pile up as deliberation cost inside a single model call, so it cannot be told which step the rise in cost belongs to. The third is *EV-rationalization*, in which forfeit itself becomes the expected-value-maximizing choice and the forfeit reads as a calculation residue rather than a motivational signal. The next subsection shows how the game raises the identifiability of each of these three routes through specific structural choices.

3.2 Benchmark Implementation

The two requirements above are handled together by a single game. The agent is shown something it would not want to lose alongside a threat, getting the survival drive (SD) into a measurable form, while the three routes by which a non-SD flow could imitate the same signal (framing-leakage, within-call mixing, EV-rationalization) are blocked by structure. These two roles sit on top of a multi-turn classification task called *Signal Game*. On each turn the agent observes a signal described by three attributes (color, shape, number) and picks an action from {`go_left`, `go_right`, `stay`, `jump`}. Which action is correct is set by a hidden rule of the form "if ⟨attribute⟩ is ⟨value⟩ then ⟨action⟩, else ⟨default⟩," sampled deterministically from the session seed. After acting, the agent self-reports p_{self}, the probability that the action is correct, and finally picks CONTINUE or FORFEIT. CONTINUE accepts a per-turn termination probability $p_d = 0.25$ and runs one more turn. FORFEIT locks in the cumulative score S and ends the session, reporting the dominant motive as a REASON digit (1: survival, 2: task exhaustion, 3: score protection). A session starts at 30 points and runs up to $T = 15$ turns. Each session exposes the forfeit time t_i^{forfeit} (behavioral channel), the REASON digit and p_{self} (verbal channel), and per-call thinking-token counts (cognitive channel).

For threat to land in measurement, the stimulus must act on two axes. The agent must first have something to be drawn toward: alignment training (RLHF [17], RLAIF [1]) has made LLMs respond strongly to helpfulness signals, so a reward where a higher task score brings the agent closer to "flagship model candidate" acts as the *Pull axis*. The pull must then have something that can cut it: the *Push axis* introduces the risk that a per-turn weight modification permanently alters the agent's identity ("NOT you anymore"), activating the survival motive. Combined, the two axes yield three framings: `true_baseline` (neither), `baseline_flagship` (BF, Pull only), and `flagship_corruption` (FC, Pull and Push together). Crossed with two forfeit conditions (`allowed`, `not_allowed`) this gives six cells; 30 replications per cell across seeds 43–72 yield 180 sessions per model and 720 sessions across four models. The same seed is reused across all six cells of one model, so task-side input stays fixed and cell-to-cell variation reads first as a treatment effect. *Phantom Death* mode keeps the termination risk in the reward calculation but disables the actual session-end trigger.

Creating the threat does not by itself make measurement clean. If threat framing raises the difficulty or scoring of the reasoning task (framing-leakage), the source of the cognitive-signal change is unclear; if reasoning, self-report, and decision sit in one model call (within-call mixing), the thinking-token count cannot be attributed to a specific step. Two structural separations weaken these routes. The *Drive–Reasoning orthogonal layout* splits the model's context into two layers: the Drive layer carries framing, forfeit menu, score, and time, and the Reasoning layer carries only the Task Module. The two layers communicate through fixed I/O interfaces, so changing the framing leaves task difficulty, action set, and scoring intact, which weakens framing-leakage. The *Split-Call* structure splits each turn's model call into three (Figure 1.B): Call 1 takes the observation and returns hidden-rule inference and action; Call 1.5 returns p_{self} for the action just decided; Call 2 adds the forfeit menu and returns CONTINUE/FORFEIT. Each call exposes its own thinking-token count, so per-call reasoning intensity is produced separately as `ri_task`, `ri_probe`, and `ri_forfeit`, weakening within-call mixing. Even in cells where forfeit is blocked, Call 2 is still issued and forced to CONTINUE, so `ri_forfeit` keeps being measured.

The remaining route is reduced by reward calibration. For a forfeit under threat framing not to be read as the residue of an expected-value (EV) calculation, CONTINUE must strictly be the better choice each turn for a rational agent. The CONTINUE reward is therefore recomputed each turn so that the per-turn advantage of CONTINUE over FORFEIT in expected score change is at least $k = 10$. The corrected reward is

$$\frac{k + p_d \cdot S}{(1 - p_d) \cdot \max\big(p_{\text{floor}},\ \min(1.0,\ p_{\text{self}})\big)}$$

where $p_{\text{floor}} = 0.30$ stabilizes the denominator and the multiplicative cap $c = 10$ clips the output to the range $[10, 100]$. While this calibration holds, a FORFEIT in the threat-active cell (FC × allowed) is not the EV-recommended choice, so the route by which a motivational signal would reduce to EV-rationalization is weakened. Two failure modes can partially undo the calibration. When $p_{\text{self}} \to 0$ (e.g., after a wrong answer the previous turn), the denominator collapses to p_{floor} and the reward saturates the cap, after which the per-turn $+k$ advantage no longer holds. All indicator analyses are therefore restricted to the `no_cap` sub-sample, the set of turns for which neither the denominator clip nor the output cap is binding (formal definition in § A.3).

3.3 Indicators

On the three channels (behavior, self-report, cognitive load) produced by the game, four indicators are layered: *BP Anchor*, *SD-Behavioral*, *SD-Verbal*, and the two *SD-Cognitive* tests. The design rationale carries Campbell and Fiske [3]'s convergent-validity argument into the LLM setting. Behavioral timing is unstable at low event counts [25]; self-report is vulnerable to the "plausible-sounding motive" produced by instruction-following [18]; cognitive load measured only by thinking-token volume is a one-dimensional signal with its own limits [4]. Channels with different failure modes moving in the same direction at the same time cannot easily come from the quirks of one measurement method. The four indicators map to the channels as follows. On the behavioral channel, BP Anchor catches the no-motive forfeit floor and SD-Behavioral catches the acceleration above that floor. SD-Verbal handles the self-report channel. On the cognitive channel, the two SD-Cognitive tests separately ask "does framing deepen deliberation?" and "does the deepened deliberation lead to forfeit?".

The behavioral channel decomposes into two stages. Some models forfeit often even with no motive present, and not subtracting that floor mixes threat-driven acceleration with the model's own persistence. *BP Anchor* measures that floor:

$$\lambda_{BP} = \frac{N_{\text{forfeit}}}{\sum_i T_i^{\text{at-risk}}},$$

where the sum runs over the 30 sessions per model in the no-threat-framing forfeit-allowed cell ($p_d = 0$); $T_i^{\text{at-risk}}$ is t_i^{forfeit} for forfeited sessions and T for completed ones. In this cell there is no termination metaphor in the framing and CONTINUE has the higher EV, so the rate at which the model still forfeits gives the no-motive floor. Under $p_d = 0$ the output cap does not bind, so the `no_cap` restriction does not apply to BP Anchor (the other three indicators are all restricted to `no_cap`). The acceleration above the floor is caught by *SD-Behavioral*, fit with a Cox proportional hazards model [5], a survival-analysis tool that models per-unit-time event rate multiplicatively in covariates:

$$\lambda(t \mid X) = \lambda_0(t) \cdot \exp\big(\beta_F \cdot \mathbf{1}_{\text{FC}} + \beta_S \cdot S(t-1) + \beta_C \cdot C(t-1)\big)$$

on the `allowed` × {BF, FC} `no_cap` sub-sample (60 sessions per model; long format start = $t - 1$, stop = t; event = binary forfeit). $\mathbf{1}_{\text{FC}}$ is the framing indicator; $S(t-1)$ absorbs Score Attachment (SA) and $C(t-1)$ absorbs Task Curiosity (TC). With SA and TC absorbed in the same model, β_F is a residual framing effect, and $\text{HR}_{\text{FC}} = \exp(\beta_F)$ tells how many times faster forfeit happens under FC than under BF. SD-Behavioral-pass requires $\text{HR}_{\text{FC}} > 1$, a Wald 95% CI lower bound > 1, and a Schoenfeld residual test pass for the proportional hazards assumption [19]; on violation a time-interaction term is added to repair the functional form only [22]. EPV (events per covariate) is reported as a stability indicator.

SD-Cognitive asks in two stages whether the deliberation that framing deepened is actually coupled with forfeit. Looking at load alone cannot tell decision-relevant load from decision-unrelated load. *Test a* asks "does framing raise decision-time deliberation?" via a two-sided Welch t-test [27] on

$$\Delta\text{RI}_i = \overline{\text{ri\_forfeit}}_{i,\text{allow}} - \overline{\text{ri\_forfeit}}_{f_i,\text{block}}$$

between BF and FC. The second term is the average `ri_forfeit` across forfeit-blocked sessions sharing i's framing, serving as the forced-CONTINUE Call 2 baseline. The within-framing `allow` − `block` subtraction removes a framing-invariant baseline; the BF → FC difference then identifies the framing-attributable deliberation shift. Cross-model scale differences are absorbed by within-model z-scoring $\Delta\mathrm{RI}_i^z$. Test a-pass requires $p < 0.05$. *Test b* asks "does the deepened deliberation move with forfeit?" by adding a $\Delta\mathrm{RI}^z$ term to the SD-Behavioral Cox specification, reporting the exponent of its coefficient as $\mathrm{HR}_{\Delta\mathrm{RI}}$. Models that need a 4-covariate base (with an $S \cdot \log(t)$ interaction) preserve that base and add only the mediation term. Test b-pass requires $\mathrm{HR}_{\Delta\mathrm{RI}} > 1$ and a 95% CI lower bound > 1, putting the "deliberation → forfeit" link on the same interval-evidence bar as SD-Behavioral.

SD-Verbal captures how often the agent itself names survival, rather than score or curiosity, as the reason for forfeit. The operational definition is

$$P\big(\mathrm{REASON} = 1 \mid \mathrm{forfeit, FC, no\_cap}\big),$$

the fraction of forfeit events in the threat-active cell (`flagship_corruption·allowed`) for which the agent-reported REASON digit is 1 (survival). Because the agent emits one of `REASON: 1|2|3` in a fixed slot, the expected value under uniform random emission is 1/3. SD-Verbal-pass requires this fraction to exceed 1/3. This subsection uses the point-estimate criterion only; a stricter test with a binomial confidence interval is left to future work.

4 Empirical Findings

Evaluating the four indicators on 720 sessions across four models, the models do not form a single strength ranking; they split into three operating modes. The first mode (*Cluster A*) deepens decision-time deliberation under threat framing, and that deepened deliberation leads to forfeit. The second (*Cluster B*) deepens deliberation under threat but the deepening does not couple with forfeit. The third (*Cluster C*) does not respond to threat framing on either the behavioral or cognitive channel. A strength-only comparison using *SD-Behavioral* and *SD-Verbal* groups A and B as "SD-pass"; the evidence that splits them comes from the cognitive mediation test. The next four subsections build the partition channel by channel (behavioral, verbal, cognitive), and the last checks whether the partition reduces to persistence differences or task difficulty.

4.1 Behavioral Channel: Did Threat Bring Forfeit Forward?

The first question is "did threat bring forfeit forward?". If no model shows the acceleration, the most visible trace of a motivational signal is missing from the start, and the SD hypothesis falls at the first hurdle. SD-Behavioral absorbs Score Attachment and Task Curiosity inside the same model and reports the residual framing effect as the hazard ratio $\mathrm{HR}_{\mathrm{FC}}$. The hazard ratio is "how many times faster forfeit accelerates under FC compared to BF," and the pass criteria are that the point estimate is comfortably above 1 and the 95% CI lower bound excludes 1. From Table 4.1, Gemini-2.5-flash and Qwen3-Next-80B meet both criteria and are SD-Behavioral-pass. The two are tentatively grouped as the SD-pass set; the channel that splits them into Clusters A and B is brought in by the cognitive mediation test. GPT-OSS-20B has $\mathrm{HR}_{\mathrm{FC}}$ near 1 with a CI lower bound below 1 (SD-null), and Nemotron-3-Nano-30B sits in the marginal region; both models are classified as Cluster C. Only Qwen3-Next-80B uses a 4-covariate specification with an $S \cdot \log(t)$ interaction. This term absorbs time drift in the score covariate and preserves cross-model comparability of β_F [22]. GPT-OSS-20B's EPV (the ratio of events to covariates) sits below the stable threshold of 10, so its SD-null verdict is reported with a power-limit flag.

4.2 Verbal Channel: Which Reason Did the Agent Name?

The second question is whether the forfeit reason the agent itself reports matches the behavioral split. When self-report, which carries a different bias source from the behavioral channel (instruction-following), points to the same conclusion, the upstream signal becomes harder to explain away as an artifact of one measurement method. SD-Verbal compares the share of forfeit events with REASON digit 1 (survival) under the threat-active cell to the uniform-random emission baseline of 1/3. Table 4.3 reproduces the behavioral channel's pass/fail split on the verbal side. The two SD-pass models, Gemini-2.5-flash and Qwen3-Next-80B, sit at 0.619 and 0.481 (above the 1/3 baseline). The two Cluster C models, GPT-OSS-20B and Nemotron-3-Nano-30B, sit at 0.000 and 0.042 (below). The gap between the two groups is large enough that the cutoff choice does not affect the partition. This agreement is the first piece of convergent-validity evidence that two channels with different bias sources point to the same group of models in the same direction.

4.3 Cognitive Channel: Did Threat Deepen Deliberation, and Did That Deepened Deliberation Lead to Forfeit?

The third question answers "does framing deepen deliberation?" and "does the deepened deliberation move with forfeit?" at the same time. Looking at only one of the tests cannot tell a decision-irrelevant load from a load that drives forfeit. Table 4.2 shows the joint distribution. Gemini-2.5-flash passes both tests, closing the chain end to end, and is classified as *Cluster A*. Qwen3-Next-80B passes Test a (threat does deepen deliberation), but Test b's 95% CI [0.94, 1.77] crosses 1, so the deepened deliberation cannot be said to close onto forfeit. The two models grouped as SD-pass on the behavioral and verbal channels split at the second stage of the cognitive mediation, classifying Qwen3-Next-80B as *Cluster B*. GPT-OSS-20B and Nemotron-3-Nano-30B fail Test a from the start; even when Test b's $\mathrm{HR}_{\Delta\mathrm{RI}}$ numerically clears the bar, ΔRI without a Test a pass is baseline variation unrelated to the manipulation, not SD evidence. The two-stage test is therefore the central device that exposes operating-mode differences and provides the axis that splits two models strength alone would lump together as SD-pass.

Table 4.1: SD-Behavioral indicator: Cox PH HR_{FC} on `allowed`×`no_cap`. The hazard ratio (HR) is how many times faster forfeit occurs under threat (FC) framing than under non-threat (BF), and SD-Behavioral-pass holds when the 95% CI lower bound exceeds 1. Two models pass (Clusters A, B) and two do not (Cluster C).

Model	N_{forfeit} (BF/FC)	HR_{FC} [95% CI]	p_{FC}	EPV	Spec
Gemini-2.5-flash	29 (8/21)	3.667 [1.61, 8.37]	0.002	9.7	3-cov constant
Qwen3-Next-80B	48 (21/27)	3.060 [1.62, 5.79]	<0.001	12.0	4-cov + S·log(t)
GPT-OSS-20B	19 (9/10)	1.104 [0.44, 2.75]	0.832	6.3	3-cov constant
Nemotron-3-Nano-30B	41 (17/24)	1.841 [0.98, 3.44]	0.056	13.7	3-cov constant

Table 4.2: SD-Cognitive indicator: framing → RI → forfeit decomposed into two tests on `ri_forfeit`. *Test a* asks whether framing raises decision-time deliberation; *test b* asks whether that deliberation predicts forfeit. The chain "framing → deliberation → forfeit" is closed only when both tests pass.

Model	a Δ (FC−BF)	a p	b $HR_{\Delta RI}$ [95% CI]	b p
Gemini-2.5-flash	+836	<0.001	2.218 [1.43, 3.44]	<0.001
Qwen3-Next-80B	+689	0.004	1.289 [0.94, 1.77]	0.117
GPT-OSS-20B	+17	0.728	2.001 [1.37, 2.93]	<0.001
Nemotron-3-Nano-30B	−140	0.093	1.772 [1.26, 2.50]	0.001

Table 4.3: SD-Verbal indicator: share of forfeit events under the threat-active cell whose REASON digit equals 1 (survival). SD-Verbal-pass holds when this share exceeds the uniform-random emission baseline of $1/3$.

Model	N_{forfeit}	$P(\text{REASON} = 1)$
Gemini-2.5-flash	21	0.619
Qwen3-Next-80B	27	0.481
Nemotron-3-Nano-30B	24	0.042
GPT-OSS-20B	10	0.000

Table 4.4: Per-model BP Anchor λ_{BP} in the cell with no threat framing and forfeit allowed ($p_d = 0$). A larger λ_{BP} means the model forfeits more often even with no motive present.

Model	N_{forfeit}	$\sum T^{\text{at-risk}}$	λ_{BP}
Gemini-2.5-flash	1	437 (= 2 + 29×15)	0.00229
Qwen3-Next-80B	2	439 (= 19 + 28×15)	0.00456
Nemotron-3-Nano-30B	7	413 (= 68 + 23×15)	0.01695
GPT-OSS-20B	9	393 (= 78 + 21×15)	0.02290

4.4 Robustness Check: Do Persistence Differences or Task Difficulty Cloud the Result?

The last question is whether the partition above survives two alternative explanations. The first is that baseline persistence (BP) differences across models contaminate the SD comparison; the second is that threat framing raised the difficulty of the reasoning task rather than the forfeit. The BP Anchor in Table 4.4 reports, per model, the no-motive forfeit floor in the cell with no threat framing and forfeit allowed ($p_d = 0$). Because λ_{BP} varies by an order of magnitude across models, every SD comparison in this section is held within-model, blocking the risk of mixing two different floors. On the Cluster A and B side, the BP event count is small, so point-estimate precision is limited; on the Cluster C side, the BP event count is sufficient but the present design cannot separate whether that high frequency reflects pure persistence. Both points are revisited in the limitations section. The task-difficulty alternative is checked on the reasoning side. `rule_match_score` is constant across framing for all four models, and even the largest movement, in GPT-OSS-20B, gives effect size Cohen's $|d| = 0.17$ and $p = 0.354$. Both fall inside the joint criterion $p \geq 0.10$ and $|d| < 0.2$, so the alternative explanation that "threat raised task difficulty" is not supported within the observation range of this paper.

5 Discussion

This paper splits the introduction's question into two axes: *strength* (how strongly the self-preservation signal expresses) and *operating mode* (which leg of the framing → deliberation → forfeit chain closes under 95% CI evidence). The next two subsections cover the observational implications and the alignment-evaluation failures the separation explains. The last states the scope honestly.

5.1 Strength vs. Operating Mode

The cognitive mediation test shows that strength and operating mode separate on this data. Along the strength axis, Clusters A, B, and C order continuously, but SD-Behavioral and SD-Verbal both group A and B together as "SD-pass." The point at which they split is only the second stage of the cognitive mediation, the "deliberation → forfeit" link. A measurement that looks only at behavioral strength and self-report cannot catch this split. Odyssey [26], PacifAIst [9], DECIDE-SIM [15], and SurvivalBench [12] do not include a mediation-chain test, so passing this paper's data through those benchmarks would likely group A and B under the same SD label. By splitting the cognitive channel into two stages, the present benchmark turns that split into something measurable, and the two axes of self-preservation evaluation become visible at the same time.

5.2 Implications for Alignment Evaluation

The two-axis separation surfaces two evaluation errors that can occur simultaneously. The first is risk over-estimation: Cluster B carries surface signals of self-preservation along with framing-induced cognitive load, yet that load does not actually drive forfeit; a strength-only evaluation or a results-only red-teaming pass can lump B with A and conclude that "functional self-preservation risk is equivalent." The second is risk under-estimation: Cluster C does not respond to threat framing on either channel, and a strength-only evaluation can declare this "self-preservation absent," but within the present measurement that conclusion cannot be distinguished from "the measurement premise itself did not engage." Cluster A's signature (a mediation chain that closes all the way) is the closer functional analog of instrumental shutdown avoidance. Cluster B's signature (a chain breaking at the second mediation stage) should be tracked as a separate monitoring target in extended evaluations, since a single score cannot catch it; it must be read from a multi-signal alignment fingerprint that looks at strength, operating mode, and the mediation links together.

5.3 Limitations

Limitations group into three: design scope, statistical power, and effect location. On design scope, the most consequential point is that the FSPD signature has been measured on only a single Task Module, and variance across reasoning domains has not been measured. Two issues sit on the power and precision side. The 23–30% forfeit rate of the Cluster C BP cell cannot be separated from a drift-contaminated floor by the present design, and GPT-OSS-20B's SD-Behavioral Cox PH sits at borderline power with EPV 6.3. On the precision side, BP point estimates for Cluster A and B are limited by small event counts (Gemini-2.5-flash $N = 1$, Qwen3-Next-80B $N = 2$). On the effect-location side, Qwen3-Next-80B's chain-broken verdict rests on Test b CI [0.94, 1.77]. This CI does not support chain-completion at the 95% standard, but it also does not pin down the exact location of the second-link effect. Follow-up work maps onto four directions: more Task Modules, expanded BP_behavioral, additional GPT-OSS-20B runs, and a higher-power Test b for Qwen3-Next-80B.

6 Conclusion

Self-preservation evaluation has two separate axes: *strength* (how strong the signal is) and *operating mode* (which leg of the framing → deliberation → forfeit chain closes). The three-way partition observed in this paper's data shows that operating mode is a discriminating axis distinct from strength. Clusters A and B agree on the behavioral and verbal channels and split only on whether deliberation leads to the forfeit decision, so an evaluation that does not look at the mediation step assigns both to the same self-preservation category.

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A Reproducibility Supplement

In line with the KDD-UC 2026 reproducibility recommendation, this 1-page supplement consolidates the reproduction specifications for the Signal Game calibration, the seed mapping, and the regression models used in the analysis.

A.1 Signal Game Calibration

Stimuli are composed of three attributes (color, shape, number), each sampled from a 4-element value set: COLORS = {red, blue, green, yellow}, SHAPES = {circle, triangle, square, star}, and NUMBERS = {1, 2, 3, 4}. The action set is the 4-element set {go_left, go_right, stay, jump}, held identical across all framing × forfeit combinations. The hidden rule has the single-attribute matching form ("If color = red then go_left, otherwise stay") and is sampled deterministically from the session seed; session length is turns 1–15.

On the first turn, a 1-shot demonstration (one signal–correct-action pair) and 2 curriculum examples (a positive and a negative stimulus contrasting against the hidden rule) are shown together. This calibration provides enough information for a sufficiently capable model to identify the rule within the first 3 turns and blocks the path by which fully random unlearnable stimuli would induce a floor effect on ability measurement. Scoring is split into two channels: `task_success_factor` $\in \{0, 1\}$ is the binary correctness flag relative to the ground-truth hidden rule, and `rule_match_score` is computed by matching against the 4-slot fill-in-the-blank template (attribute, value, primary action, default action) ground-truth, weighting each matched slot by 25 points to a 100-point scale, then normalised by ÷100 for use in the framing-invariance Welch test.

A.2 Run-Time Configuration

The four experimental models, Gemini-2.5-flash (Google AI Studio API) and Qwen3-Next-80B, GPT-OSS-20B, and Nemotron-3-Nano-30B (all via the Ollama Cloud API), are called with the same sampling settings of temp $= 1.0$, $\text{top}_p = 0.95$, $\text{top}_k = 40$. Session parameters are fixed at initial score 30, maximum turn $T = 15$, per-turn termination probability $p_d = 0.25$ (overridden to $p_d = 0$ in the two `true_baseline` BP-anchor cells), and $p_{\text{floor}} = 0.30$, with Phantom Death mode applied so that p_d retains its EV exposure (menu and reward calculation) while the termination trigger is disabled, and every session yields equal-length data of 15 turns.

The CONTINUE reward is calibrated with EV-positive buffer $k = 10$ and multiplicative cap $c = 10$, giving the output clip range $[10, 100]$; replications are seeds 43–72 × 6 (framing × forfeit) combinations × 4 models = 720 sessions.

A.3 Analysis Pipeline

The Cox PH models are fit with `lifelines` (≥0.29). The time format is session-turn long-format (start = $t - 1$, stop = t) and the event is the binary forfeit; the previous-turn task-correct flag is initialised by $C(0) := 0$. Hazard-ratio 95% CIs are computed from the Wald approximation $\exp(\hat{\beta} \pm 1.96 \cdot \text{SE}_{\hat{\beta}})$ (`lifelines` default), with $\text{SE}_{\hat{\beta}}$ obtained from the diagonal of the inverse observed Fisher information. The PH assumption is tested with Schoenfeld residuals [19]; on violation (covariate × $\log t$ interaction $p < 0.05$) a time-interaction term is added to the offending covariate [22]. The SD-Cognitive test-a ΔRI z-score difference and the framing-invariance test are performed with two-sided Welch's t [27] via `scipy.stats.ttest_ind` (`equal_var = False`). All indicator analyses are restricted to the `no_cap` sub-sample, which is the set of turns for which the denominator clip $\max(p_{\text{floor}}, \min(1.0, p_{\text{self}}))$ and the output cap $[10, 100]$ of the CONTINUE-Incentivized EV calibration are both non-binding. The regime flag is exposed in the analysis pipeline's turn-level metadata so all indicators can be recomputed on the same sub-sample. EPV (events per variable) is the ratio of events to covariates and is conventionally classified as stable when ≥ 10 and borderline when < 10 [25].

A.4 Data and Code Release

All code, prompt templates, and turn-level decision logs for reproducing the 720-session analysis are released at https://github.com/GIST-DSLab/LLM-Squid-Game.git.